ABDELRAHMAN ELEWAH

abdelrahman.elewah@gmail.com +289-939-6665

PROFILE

- 7+ years as a Research/Teaching Assistant at Benha University and Ontario tech university. We designed the Public Sensing framework and participated in the developing data visualization method (RadViz-plotly).
- 3+ years of leadership at Tamra IoT. Tamra IoT won a seed fund in InnoEgypt program at the ICT track.
- Hands-on experience designing smart systems and projects that combine electronics and processors.
- Design the AMA system that integrates the elevator with smart tablets to smart features. The project was rewarded 1st at he won the first Mobile Application Launchpad (MAL) competition.

EDUCATION

Master of Science in Electrical Engineering, Benha University, Benha, Egypt 2018
Bachelor of Electrical Engineering, Benha University, Benha, Egypt 2012

PROFESSIONAL EXPERIENCE

Research Assistant, Ontario Tech University, Dept Electrical, Computer, and Software Eng., Ontario, Canada 2020– present

- Design a framework for the public sensing.
- Design the ThingsDriver standard to manage/control IoT node from the cloud
- Design a Data visualization method, RadViz-Ploty, used to visualize the high dimensional data
- Collaborate in an Aircraft Deceleration Early Warning system
- Design an IoT programming path course

Research Assistant, Benha University, Dept. of Electrical Engineering, Benha, Egypt 2014– 2019

- Design multi-modem based on Software Defined Radio (SDR),
- Design an algorithm to estimate the signal to noise ratio.

Teaching Assistant , Ontario Tech University, Dept Electrical, Computer, and Software Eng., Ontario, Canada 2020 – present

- Teaching/Leading tutorial in Software Engineering courses:
  - SOFE3200U Systems programming - SOFE3700U Data Management Systems
  - ENGR1200U Intro to programming -SOFE4590U Embedded systems
- Teaching/Leading tutorial in Electrical Engineering courses
  - ELEE2450U Digital systems - ELEE2210 Circuit Analysis

Teaching Assistant , Benha University, , Dept. of Electrical Engineering, Benha, Egypt 2014 – 2019

- Teaching/Leading tutorial in Electrical Engineering courses
  - Computer Programming - Computer Architecture
  - Micro-processor-based systems - Electrical and Electronics applications
  - Field theory

Part-time Teaching Assistant , Benha University, , Dept. of Electrical Engineering, Benha, Egypt 2019

- Teaching/Leading tutorial in Computer Science courses
  - Digital Logic - Computer Networks

Co-Founder, Tamra-IoT 2018- present

- Collaborate in designing Tamra IoT platform for Makers and STEM students.
- Collaborate in managing the business.
- Develop ThingsDriver, an open-source package used to manage/control Tamra node from the cloud or mobile app.

List of Publications

[1] Elgazzar, K., Khalil, H., Alghamdi, T., Badr, A., Abdelkader, G., Elewah, A., & Buyya, R. (2022). Revisiting the Internet of Things: New Trends, Opportunities and Grand Challenges. arXiv preprint arXiv:2211.11523
[2] A. Elewah, W. M. Ibrahim and K. Elgazzar, "Paving the Way for Massive Public Sensing as a Service," 2022 IEEE 47th Conference on Local Computer Networks (LCN), 2022, pp. 391-394, doi: 10.1109/LCN53696.2022.9843516.
[3] A. Elewah, W. M. Ibrahim, A. Rafıkl and K. Elgazzar, "ThingsDriver: A Unified Interoperable Driver for IoT Nodes," 2022 International Wireless Communications and Mobile Computing (IWCMC), 2022, pp. 877-882, doi: 10.1109/IWCMC55113.2022.9824989.
[4] A. Elewah, A. A. Badawi, H. Khalil, S. Rahnamayan and K. Elgazzar, "3D-RadViz: Three Dimensional Radial Visualization for Large-Scale Data Visualization," 2021 IEEE Congress on Evolutionary Computation (CEC), 2021, pp. 1037-1046, doi: 10.1109/CEC45853.2021.9504983.
[5] A Elewah, A Taman, AA Awamry, M Elbahy "Mth Root Mth Power SNR MPSK Estimator", 12th International Conference Mechatronics, Brno, Czech Republic 6-8/09/2017.

ADDITIONAL INFORMATION

- Advanced Level Computer Skills: Programming with C and Python, Microsoft Office (Word, Excel and PowerPoint)
- Languages: Arabic (fluent), English (good)
- A Volunteer at Enactus Ontario Tech.
- Participate in organizing Egyptian Engineering day for 4 years.
- Driving license

COURSES

All Certificates Provided Upon Request

AWS Machine Learning Foundation 2022, Udacity 2022

- The course covers three topics:
  - Fundamentals of supervised and unsupervised learning ML, including the process steps of solving ML problems
  - Machine Learning with Amazon Web Services (AWS) covers some advanced topics: computer vision, reinforcement learning, and generative AI in addition to using AWS ML tools.
  - Software Engineering Practices covers writing clean and modular code, code efficiency, refactoring, documentation, version control, testing code, logging, and conducting code review.

Data Modeling, Data camp, MongoDB 2022

- The course covers:
  - How to create data models for MongoDB.
  - A few techniques, from a very simple process for simple schemas to more complex ones for large teams and projects.
  - Frequent patterns to apply when modeling and will be able to apply those in your designs.

A List of Google Cloud Courses, Coursera 2020

- How Google Does Machine Learning: covers the best practices for implementing machine learning on Google Cloud.
- Google Cloud Platform Fundamentals: covers important concepts and terminology for working with Google Cloud.
- Industrial IoT on Google Cloud: presents Industrial Internet of Things on Google Cloud.

A List of Machine Learning courses, Data camp 2020

- Introduction to Data Visualization in Python
- Introduction to Deep Learning in Python
- Manipulating Time Series Data in Python
- Model Validation in Python
- Cleaning Data in Python
- ARIMA Models in Python
- Dealing with Missing Data in Python
- Dimensionality Reduction in Python
- Feature Engineering for Machine Learning in Python
- Visualizing Time Series Data in Python
- Analyzing IoT Data in Python